

Daerhan Jianati

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Education

University of Illinois Urbana-Champaign

Aug. 2026 – May 2028

School of Information Sciences

M.S. in Information Management

Nanjing University

Sep. 2022 – Jun. 2026

School of Electronic Science and Engineering

B.S. in Electronic Information Science & Technology

Honor: Ethnic Minority Students' Pomegranate Seed Scholarship (two-time recipient)

Research Interests

Computational Imaging · Multispectral Imaging · Statistical Image Reconstruction

Multimodal Sensing and Perception · Human-Perception-Informed Imaging

Selected Research Experience

Self-Supervised RGB–Active NIR Fusion for Visual Perception

Nov. 2025 – Present

Project Lead and First Author; Advisor: G. Lu

- Developed a lightweight single-channel fusion framework for pixel-aligned RGB and active NIR using pixel-wise mixing, stereo reprojection, and an exposure-aware reliability gate.
- Evaluated the fused representation across five major downstream tasks: object detection, stereo depth estimation, feature matching, structure-from-motion (SfM), and simultaneous localization and mapping (SLAM), while also measuring runtime efficiency, reaching 79.36 FPS with 560 MB peak GPU memory.
- Currently extending the study with expanded training and evaluation protocols, robustness analysis under calibration perturbations, ablation studies, and experiments under challenging illumination conditions.

Per-Pixel Illumination Estimation from HR-RGB and LR-HSI

Jun. 2025 – Oct. 2025

Student Researcher, CITE Lab, Nanjing University

- Developed a multi-sensor pipeline combining high-resolution RGB and low-resolution hyperspectral measurements for per-pixel illumination estimation.
- Reproduced baseline results and evaluated robustness across multiple spatial downsampling scales and fusion settings.

Unitree Go2 Multi-Floor Guide Navigation

Apr. 2026 – Jul. 2026

Research Assistant, Human-Computer Interaction Lab, Nanjing University

Links: [Project](#) | [Documentation](#)

- Integrated a ROS 2/Nav2 navigation pipeline for a Unitree Go2 EDU, including LiDAR processing, mapping, localization, global planning, path following, and Unitree motion-command integration.
- Developed semantic destinations, robot-safe anchors, multi-floor map organization, planner-validation tools, and a human-assisted elevator workflow.
- Open-sourced bilingual documentation, maps, configurations, scripts, and debugging procedures.

Technical Skills

Programming	Python, C, MATLAB
ML & Vision Libraries	PyTorch, OpenCV, NumPy
Imaging & Vision	Multimodal Fusion, Object Detection, Stereo Depth Estimation, Feature Matching, SfM/SLAM Evaluation, Illumination Estimation, Sensor Calibration
Robotics	ROS 2, Robotic Navigation, SLAM, Unitree Go2
Sensing Modalities	RGB, Active NIR, Hyperspectral, Polarization, LiDAR
Tools & Platforms	Linux, Git, LaTeX, Arduino, VMware

Additional Research & Projects

Exploratory RGB/NIR/Polarization Depth Estimation

Aug. 2025 – Oct. 2025

Explored multimodal fusion, custom losses, knowledge distillation, calibration, and sparse LiDAR supervision.

3D Dynamic Heart Reconstruction from 2D Medical Imaging

Feb. 2025 – May 2025

Project Lead, Course Project; led preprocessing, segmentation, reconstruction, and myocardial-motion visualization.

Intelligent Risk Butler: Exchange-Rate Risk Inspection System

Jan. 2025 – Jun. 2025

Core Member, Algorithm Group, 20th Citi Cup; developed an LLM module and exchange-rate risk signal pipeline.

Quantum Implementation of Finite Element Analysis

Jan. 2025 – Feb. 2025

Exchange Project, North Carolina State University; implemented block-encoding/QSP-based solvers and explored BPX preconditioning.

Reservoir-Computing-Transistor-Based Fall Detection

Nov. 2023 – Dec. 2024

National Undergraduate Innovation Program; contributed to a hardware-software system integrating analog computing, Arduino, Bluetooth, and vision-based motion analysis.

Industry Experience

Data Analysis Intern

Jul. 2024 – Sep. 2024

College of Water Conservancy and Architectural Engineering, Shihezi University

- Processed multivariate environmental sensor data and evaluated machine-learning models for anomaly detection in irrigation leakage and drought stress.

Research Outputs

Manuscript

Jun. 2026 – Present

Daerhan Jianati and G. Lu. “Learning a Single-Channel Fusion Representation from RGB and Active NIR via Self-Supervision.”

Patent

Patent Application

Jan. 2025

“A System and Method for Elderly Fall Detection Based on Reservoir Computing Transistors.”
Fifth inventor; Application No. CN202411352818.X.